

Proportional Control for Maneuvering Flapping Flight

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June 5, 2026

1 Model Description

Table 1 lists the nondimensionalized parameters in the dynamic model. Ranges for the parameters L_w^* , A_w^*/m^* , and ω^* were obtained from the morphological data in [1] and the wingbeat frequency data in [2] for *Anisoptera*.

Table 1: Dimensionless Parameters

Parameter	Description	Range
$L_w^* = \frac{L_w}{L}$	Wing length ratio	[0.65, 0.85]
$A_w^*/m^* = \frac{A_w \rho_{\text{air}} L}{m}$	Inverse wing loading (per wing)	[0.05, 0.25]
$\omega^* = \omega \sqrt{\frac{L}{g}}$	Wing beat frequency	[8, 20]
γ_0	Wing stroke plane angle	[-180°, 180°]
ϕ_1	Wing flapping amplitude	[0°, 60°]
ψ_0	Mean wing pitch angle	[-60°, 60°]
ψ_1	Wing pitch amplitude	[0°, 60°]
δ_0	Wing pitch phase	[-180°, 180°]
σ_0	Fore/hind phase	[-180°, 180°]

Lift and drag coefficients are given by

$$C_L = C_{L,0} \sin 2\alpha$$

$$C_D = C_{D,0} \cos^2 \alpha + C_{D,\frac{\pi}{2}} \sin^2 \alpha$$

with $C_{L,0} = 1.5$, $C_{D,0} = 0.1$, $C_{D,\frac{\pi}{2}} = 2$

2 Equilibrium Solution Space

A good first step might be to identify those parameter values that give an average aerodynamic force over a cycle that is equal to or greater than the weight, i.e. $F_a^* = \frac{F_a}{m_b g} \geq 1$. Outside of this group of solutions, hovering and meaningful maneuvering flight cannot be achieved. This is independent of the stroke plane angle γ_0 and the fore/hind phase shift σ_0 , which limits us to just 7 parameters. We can further eliminate one parameter by collapsing L_w^* and ϕ_1 into a single wing tip displacement amplitude parameter $s_0^* = \frac{2}{3} L_w^* \phi_1$, resulting in 6 morphological and wingbeat parameters.

As a first, quick verification we compare the $F_a^* = 1$ contour result in the $(A^*/m^*, \omega^*)$ plane to the simplified analytical result of [3] for a translating element with constant 45° pitch through each beat half:

$$(F_a^*)_{\text{constant pitch}} = \frac{C_{L,0}}{4} \frac{A^*}{m^*} (s_0^* \omega^*)^2 \quad (1)$$

Note $A^* = 4A_w^*$. The result is obtained for $s_0^* = 1.5$ (achieved in practice with a small ϕ_1 and a large L_w^*), $\psi_0 = 0^\circ$, $\psi_1 = 45^\circ$, and $\delta_0 = 90^\circ$. Crucially we do this assuming zero body velocity: the velocity of the wings relative to the air is equal to the velocity of the wings relative to the body. Contours are shown in Figure 1. Using a single wing element at the $\frac{2}{3}$ -span station, there is very close agreement, as expected, with a small difference attributable to the sinusoidal wing pitch (as opposed to a square wave). The wing element spends less time at large angle of attack so the mean force is slightly lower. As the number of elements is increased F_a^* converges to a lower value (by about 25%), causing the contour to shift towards higher A^*/m^* and ω^* .

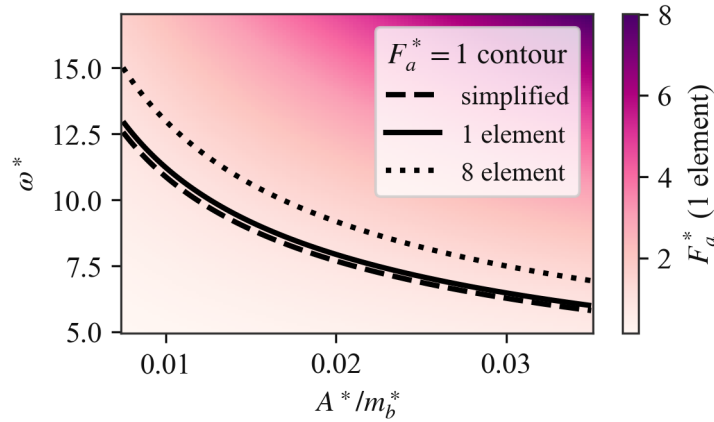


Figure 1: Verification of $F_a^* = 1$ contour at reference condition

The present model allows for more complex wing motions than a simple symmetric stroke: the mean pitching angle ψ_0 may be nonzero, and the phase shift δ_0 between the pitching and flapping motions may be different from 90° . One question we can ask is whether the contribution of the pitch to the force magnitude F_a^* can be represented as a coefficient that would depend on $(\psi_0, \psi_1, \delta_0)$, independent of the other 3 parameters. Looking back to the simple analytical expression for F_a^* for constant pitch, we can extract the dimensionless dynamic pressure

$$q^* = \frac{1}{4} \frac{A^*}{m^*} (s_0^* \omega^*)^2 = \frac{A_w^*}{m^*} (s_0^* \omega^*)^2 \quad (2)$$

We can then evaluate F_a^*/q^* and check whether it is independent of $(\frac{A_w^*}{m^*}, s_0^*, \omega^*)$ over the parameter space, i.e. whether the two groups of variables can be cleanly separated. Again, we do this first assuming zero body velocity, such that $F_a^* = 1$, applied in the right direction, would result in a hover. As can be seen in Figure 2, the separation is clean, with small amounts of scatter associated with the nonlinearity of the flapping motion: at large wing angles, there is an appreciable change in the effective angle of attack, but the effect is small. The right side of the plot is obtained by picking a reference amplitude ($s_0^* = 0.5$) to evaluate F_a^* . The result is completely independent of A_w^*/m^* and ω^* because they appear purely as multiplicative factors in the analytical expression. It might be clearer to do a purely analytical treatment of this part.

This decomposition appears to be useful because C_ψ provides a measure of the power efficiency of the wingbeat. A given net force can be achieved with a faster or slower wingbeat depending on the wing pitch

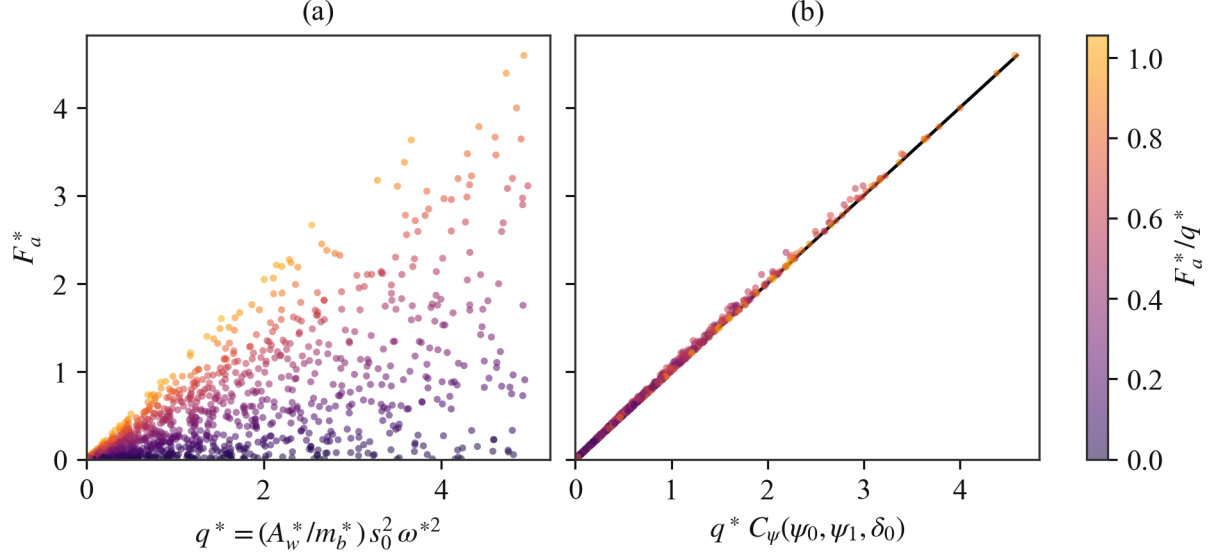


Figure 2: Separation of dynamic pressure and wing pitch factors of the aerodynamic force at zero body velocity. (a) Scatter of mean aerodynamic force as a function of dynamic pressure for random sampling of parameter space (b) Collapse of scatter by introduction of C_ψ parameter evaluated as F_a^*/q^* for a reference amplitude value $s_0^* = 0.5$

pattern. If the pattern is such that C_ψ is high, then the wing doesn't have to beat as fast, and less energy is expended (provided any additional energy required to impart that particular pitch motion does not cancel out this benefit, which seems like more of an edge case). Presumably then, values of ψ_0 , ψ_1 , and δ_0 that result in higher C_ψ would be preferred in nature. Looking again at the simplified case from earlier it seems the optimum should sit near $\psi_0 = 0$, $\psi_1 = 45^\circ$, $\delta_0 = 90^\circ$, since a 45° angle of attack maximizes lift (limit case for square wave pitch: $C_\psi = C_{L,0}$)

Since we are interested not just in hover but in maneuvering flight, we of course need to consider the case where the body velocity is nonzero.

References

- [1] J. M. Wakeling. Odonatan wing and body morphologies. *Odonatologica*, 26(1):35–52, 1997. URL <https://natuurtijdschriften.nl/pub/592186>.
- [2] C. Aracheloff, R. Garrouste, A. Nel, R. Godoy-Diana, and B. Thiria. Subtle frequency matching reveals resonant phenomenon in the flight of Odonata. *Journal of The Royal Society Interface*, 21(219), 2024. doi: 10.1098/rsif.2024.0401.
- [3] Owen C. Wetherbee and Z. Jane Wang. Stable flapping flight in morphological space: Model, simulation, and explicit stability criteria. *Proceedings of the National Academy of Sciences*, 123(18):e2533138123, 2026. doi: 10.1073/pnas.2533138123. URL <https://www.pnas.org/doi/abs/10.1073/pnas.2533138123>.